

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk: **2-FENOKSIETANOL**  
 Product Description: **2-Phenoxyethanol**  
 Cat No. : 130240000; 130240010; 130240025; 130240051; 130240250; 130242500  
 Synonyms Ethylene glycol monophenyl ether  
 CAS No 122-99-6  
 Molecular Formula C8 H10 O2

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

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 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                   |
|----------------------------------------------------|-------------------|
| Acute oral toxicity                                | Category 4 (H302) |
| Serious Eye Damage/Eye Irritation                  | Category 1 (H318) |
| Specific target organ toxicity - (single exposure) | Category 3 (H335) |

**Label Elements**


Signal Word

Danger

**Hazard Statements**

H302 - Harmful if swallowed

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H318 - Causes serious eye damage  
H335 - May cause respiratory irritation

## Precautionary Statements

### Prevention

P261 - Avoid breathing dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                        | CAS No   | Weight % |
|----------------------------------|----------|----------|
| Ethylene glycol monophenyl ether | 122-99-6 | <= 100   |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes severe eye damage.

### Indication of any immediate medical attention and special treatment needed

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## Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors.

### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental precautions

Should not be released into the environment.

### Methods and Material for Containment and Cleaning Up

Keep in suitable, closed containers for disposal. Soak up with inert absorbent material.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

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## Control Parameters

| Component                        | European Union | The United Kingdom | Germany                                                                                                                                                                                                                                                                                                                                                  |
|----------------------------------|----------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol monophenyl ether |                |                    | TWA: 1 ppm (8 Stunden). AGW - exposure factor 1<br>TWA: 5.7 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 1<br>TWA: 1 ppm (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>TWA: 5.7 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 1 ppm<br>Höhepunkt: 5.7 mg/m <sup>3</sup> |

## Exposure Controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### **Respiratory Protection**

No protective equipment is needed under normal use conditions

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### **Appearance**

Colorless

#### **Physical State**

Liquid

#### **Odor**

aromatic

#### **Odor Threshold**

No data available

#### **pH**

8-9

10g/100ml water/MeOH

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|                     |                             |                                   |
|---------------------|-----------------------------|-----------------------------------|
| Melting Point/Range | 11 - 13 °C / 51.8 - 55.4 °F |                                   |
| Softening Point     | No data available           |                                   |
| Boiling Point/Range | 237 °C / 458.6 °F           | @ 760 mmHg                        |
| Flash Point         | 130 °C / 266 °F             | Method - No information available |

|                          |                   |        |
|--------------------------|-------------------|--------|
| Evaporation Rate         | No data available |        |
| Flammability (solid,gas) | Not applicable    | Liquid |
| Explosion Limits         | No data available |        |

|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | 0.04 mbar @ 20 °C        |             |
| Vapor Density                | 4.8 (Air = 1.0)          | (Air = 1.0) |
| Specific Gravity / Density   | 1.100                    |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | 30 g/l (20°C)            |             |
| Solubility in other solvents | No information available |             |

|                                         |         |
|-----------------------------------------|---------|
| Partition Coefficient (n-octanol/water) |         |
| Component                               | log Pow |
| Ethylene glycol monophenyl ether        | 1.107   |
|                                         | 1.2     |

|                           |                          |
|---------------------------|--------------------------|
| Autoignition Temperature  | 430 °C / 806 °F          |
| Decomposition Temperature | No data available        |
| Viscosity                 | 29 mPa s at 20 °C        |
| Explosive Properties      | No information available |
| Oxidizing Properties      | No information available |

|                   |           |
|-------------------|-----------|
| Molecular Formula | C8 H10 O2 |
| Molecular Weight  | 138.17    |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under recommended storage conditions.

### Possibility of Hazardous Reactions

|                          |                                          |
|--------------------------|------------------------------------------|
| Hazardous Polymerization | Hazardous polymerization does not occur. |
| Hazardous Reactions      | None under normal processing.            |

### Conditions to Avoid

Incompatible products.

### Incompatible Materials

Strong oxidizing agents. Acid anhydrides. Acid chlorides.

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## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component                        | LD50 Oral                 | LD50 Dermal               | LC50 Inhalation               |
|----------------------------------|---------------------------|---------------------------|-------------------------------|
| Ethylene glycol monophenyl ether | LD50 = 1850 mg/kg ( Rat ) | LD50 = 5 mL/kg ( Rabbit ) | LC50 > 0.057 mg/L ( Rat ) 8 h |

| Component                        | ECHA (RAC) ATE (Oral) | ECHA (RAC) ATE (Dermal) | ECHA (RAC) ATE (Inhalation) |
|----------------------------------|-----------------------|-------------------------|-----------------------------|
| Ethylene glycol monophenyl ether | ATE = 1394 mg/kg bw   | -                       | -                           |

ECHA (RAC) - Committee for Risk Assessment - European CHemicals Agency  
ATE - Acute Toxicity Estimate; mg/kg bw - milligrams per kilogram of body weight

##### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

##### (c) serious eye damage/irritation;

Category 1

##### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

##### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

##### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

##### (g) reproductive toxicity;

Based on available data, the classification criteria are not met

##### (h) STOT-single exposure;

Category 3

Results / Target organs

Respiratory system.

##### (i) STOT-repeated exposure;

Based on available data, the classification criteria are not met

Target Organs

None known.

##### (j) aspiration hazard;

Based on available data, the classification criteria are not met

#### Other Adverse Effects

Symptoms / effects, both acute and No information available.

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delayed

Endocrine Disrupting Properties

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects

Do not empty into drains. .

| Component                        | Freshwater Fish                                                                                                    | Water Flea                            | Freshwater Algae                                | Microtox                                       |
|----------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------|-------------------------------------------------|------------------------------------------------|
| Ethylene glycol monophenyl ether | LC50: = 366 mg/L, 96h static (Pimephales promelas)<br>LC50: 337 - 352 mg/L, 96h flow-through (Pimephales promelas) | EC50: > 500 mg/L, 48h (Daphnia magna) | EC50: > 500 mg/L, 72h (Desmodesmus subspicatus) | EC50 = 32.4 mg/L 5 min<br>EC50 = 880 mg/L 17 h |

Persistence and degradability

Persistence

Persistence is unlikely.

Bioaccumulative potential

Bioaccumulation is unlikely

| Component                        | log Pow      | Bioconcentration factor (BCF) |
|----------------------------------|--------------|-------------------------------|
| Ethylene glycol monophenyl ether | 1.107<br>1.2 | No data available             |

Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

Waste treatment methods

**Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

**Other Information**

Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer

## SECTION 14: TRANSPORT INFORMATION

IMDG/IMO

Not regulated

Road and Rail Transport

Not regulated

IATA

Not regulated

ACR13024

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**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component                        | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|----------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Ethylene glycol monophenyl ether | 204-589-7 | X    | X   | X     | X    | X    | X     | X    | KE-28257 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date** 21-Mar-2025  
**Revision Summary** SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health**



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## Disclaimer

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**End of Safety Data Sheet**